Sina Masoumi Shahrbabak

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Summary

Biomedical Engineering researcher focused on machine learning for physiological time-series and multimodal clinical data. Experience developing data-driven models for cardiovascular health using ECG, PPG, and ABP signals. Strong background in Python/PyTorch with publications in biomedical engineering venues.

Education

PhD student, Mechanical Engineering, University of Maryland — *Expected Dec 2027*

Research assistant focusing on monitoring, diagnosis, and tracking of cardiovascular disease

Master of Science, Mechanical Engineering, University of Maryland — *May 2025*, GPA: 3.9

Notable Courses: Computational Statistics, Machine Learning: Theory and Applications, Applied Machine Learning for Engineering, Optimal Estimation of Dynamical Systems

Bachelor of Science, Mechanical Engineering, Sharif University of Technology — *2022*, GPA: 3.9

Technical Skills

Programming Languages: Python (Advanced), R, MATLAB

Deep Learning & ML Frameworks: PyTorch, scikit-learn, NumPy

Methodologies & Algorithms: Time-Series Analysis, Convolutional Neural Networks (CNNs), Bayesian Inference & MCMC, Generative Modeling, Signal Processing

Research Experience

- **Researcher** at University of Maryland

Supervisor: **Dr. Jin-Oh Hahn**, Professor, Department of Mechanical Engineering

Detection of Peripheral Artery Disease Using Blood Pressure Waveform Analysis

- Generated blood pressure (BP) waveforms for +100k virtual patients using a 55-compartment model of the arterial system
- Developed a 1-D CNN to diagnose peripheral artery disease using blood pressure waveforms from two peripheral sites

Development and Evaluation of Hemodynamic Monitoring Algorithms

- Conducted in-vivo experiments on +15 swine to acquire a cardiovascular waveform database (+100k cardiac cycles) of aortic blood flow and BP, ECG, PPG, and seismocardiogram (SCG)
- Implemented an ML framework using multimodal waveform features for estimating stroke volume with <10% MAPE using only wearable signals

Wearable Physiological Sensing Enabled Life Vest to Improve Mass Casualty at Sea Survivability

- Formulated a mathematical model of the left ventricle and arterial system to generate the ballistocardiogram signal from two different physiological perspectives

Predictive Modeling of Blood Pressure During Hemorrhage and Volume Resuscitation

- Established a Bayesian model for estimating nurse charting errors using noisy blood pressure traces

Enabling Closed-Loop Controlled Mitigation of the Physiological Response to Acute Mental Stress

- Quantified the sympathetic arousal during acute mental stress in 19 human subjects by leveraging wearable measurements from ECG, PPG, and SCG

Published Papers

- **Masoumi Shahrbabak, S.,** Youn, B. D., Cheng, H. M., Chen, C. H., Sung, S. H., Mukkamala, R., & Hahn, J. O. (2025). Deep Learning-Enabled Diagnosis of Abdominal Aortic Aneurysm Using Pulse Volume Recording Waveforms: An In-Silico Study. Sensors.

- **Masoumi Shahrbabak, S**., Mousavi, A., Mukkamala, R., & Hahn, J. O. (2025). In Silico Investigation of a Mathematical Model Relating the Ballistocardiogram to Aortic Blood Pressure. IEEE Transactions on Biomedical Engineering.
- **Masoumi Shahrbabak, S**., Kim, S., Youn, B. D., Cheng, H. M., Chen, C. H., Mukkamala, R., & Hahn, J. O. (2024). Peripheral Artery Disease Diagnosis Based on Deep Learning-Enabled Analysis of Non-Invasive Arterial Pulse Waveforms. Computers in Biology and Medicine.
- Zhou, Y., **Masoumi Shahrbabak, S**., Rezaei, P., Bahrami, R., Rahman, F. N., Sanchez-Perez, J. A., Gazi, A. H., Inan, O. T., & Hahn, J. O. (2026). A Virtual Experiment Generator to Replicate Cardiovascular Responses to Acute Mental Stress and Transcutaneous Median Nerve Stimulation. Journal of Dynamic Systems, Measurement, and Control.

Conference Presentations

- **Workshop Speaker**: "Digital Health Technologies and AI for Studying and Predicting Episodic Health Events." The IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI), 2025.
- **Poster Presentation**: "Peripheral Artery Disease Alters Ballistocardiogram Morphology" & "Cardiovascular Responses to Cold Water Immersion and Rewarming." *The Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2024.
- **Poster Presentation**: "In Silico Validation of a Ballistocardiogram Mathematical Model." I*EEE-EMBS International Conference on Body Sensor Networks (BSN),* 2024.
- **Poster Presentation:** "Peripheral Artery Disease Diagnosis Based on Deep Learning-Enabled Analysis of Non-Invasive Arterial Pulse Waveforms." *BHI,* 2023.